

Mario A. Camarena

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EDUCATION

The University of Texas Rio Grande Valley

Master of Science, Computer Science | GPA: 4.0/4.0

Bachelor of Science, Computer Science | GPA: 3.4/4.0

Edinburg, TX

Jun 2025 – Expected 2027

May 2021 – May 2025

EXPERIENCE

Artificial Intelligence Research Assistant

Feb 2024 – Present

UTRGV MECIS (NSF CREST), Dept. of Civil Engineering

Edinburg, TX

- Independently lead experiment design, ablations, and implementation under advisor guidance; produce reproducible results.
- Survey literature to drive model/dataset decisions; present at lab meetings and conferences; first/co-first author on publications.

NASA-funded M.S. Thesis — AAM Flight Security Risk Modeling | Python, PyTorch, Pandas, NumPy, SciPy, OpenSky API

- Designed an event-driven, continuous-time flight graph from flight data + accident data, modeling aircraft as nodes and directed proximity events as edges; trained a temporal GNN to score pairwise accident risk.
- Built a synthetic collision/near-miss pipeline calibrated to real ADS-B statistics with stochastic noise; aligned **15** NTSB conflicts to OpenSky tracks (~**90%+** match rate) and validated collision geometry.

AD-SAM — Autonomous-Driving Urban-Scene Segmentation | Python, PyTorch, CUDA, Segment Anything (SAM)

- Improved urban-scene segmentation accuracy by **29%** for autonomous-driving use cases by leading a custom model dual-encoder effort from dataset prep to evaluation; first co-author, pending IEEE-T-ITS publication.
- Reduced boundary errors and class-imbalance misses by adopting a deformable decoder and hybrid loss (Focal, Dice, Lovász, Surface), improving per-class consistency across 19 classes.

Machine Learning Intern (REU)

Jun 2024 – Aug 2024

University of California, Riverside

Riverside, CA

- Benchmarked segmentation models (SAM, G-SAM, DeepLabV3, UNET, YOLO-based baselines) on real-world data and summarized trade-offs to guide model selection in autonomous transportation applications.
- Delivered an end-to-end evaluation pipeline automating inference, mIoU metric computation, and result logging; co-documented findings that informed a publication with a Ph.D. mentor.

PROJECTS

STABLES — Simulated Parking Management App | Flutter, Dart, Node.js, Express, PostgreSQL

Project Lead & Lead Developer | Live: stables-utrgv-parking-app.web.app

- Shipped a cross-platform app managing **800+** parking spots across 3 lots with live availability and reservations; led a 4-person team with code review standards and task ownership.
- Built the backend (Node.js/Express + PostgreSQL) with spatial checks and double-booking prevention; created a Python simulator generating synthetic data for load testing.

Coffee Brewing Outlier Detection — HackResearch 2025 Winner | Python, Pandas, NumPy, Matplotlib

Hackathon Team Lead | GitHub: mariocamarena/coffee_outlier

- Won HackResearch 2025 (24-hour hackathon, judged on methodology) by detecting an anomalous espresso shot from 13 unlabeled brewing logs via feature extraction and distance-based scoring; delivered a research paper under deadline pressure.
- Built an end-to-end time-series pipeline and extracted features to implement distance-based outlier scoring.

Online Portfolio — Interactive Website | Next.js, React, TypeScript, Tailwind CSS, PostgreSQL, Node.js

Solo Developer | Live: mariocam.cc

- Designed a terminal-inspired ASCII visual language and built a full-stack contact system with PostgreSQL, a protected admin dashboard, and real-time submission analytics.
- Built responsive animations (typewriter sequences, particle canvas, scroll effects) optimized for mobile and accessibility.

Custom PC Builds and Mobile Device Repair | Hardware Assembly & Troubleshooting

Freelance Hardware Technician

- Delivered custom PCs for gamers, students, and professionals; handled component selection, system tuning, cooling design, cost estimates, and ongoing client communication.
- Diagnosed and repaired **30+** smartphones end-to-end, self-sourcing parts for screen replacements and battery swaps.

SKILLS

Languages: Python; JavaScript/TypeScript; Dart; SQL; C++; Rust.

ML/CV: PyTorch; CUDA; Pandas; NumPy; Matplotlib; Scikit-learn.

Web/App: React; Next.js; Flutter; Node.js/Express; Tailwind CSS; Framer Motion.

Tools & Data: PostgreSQL; SQLite; Git/GitHub; Linux; Jupyter; REST APIs.